

Code:

```
#include <stdio.h> #include <string.h> #include <ctype.h>

#define MAXP 50 #define BUFLLEN 110

char *gems[] = { "NONE", "Garnet", "Amethyst", "Aquamarine", "Diamond", "Emerald",
"Pearl", "Ruby", "Peridot", "Sapphire", "Tourmaline", "Topaz", "Lapis", 0 };

char ponies[MAXP][BUFLLEN]; int gem_rank[MAXP];

int get_priority(char *name) { int max_p = 0; for (int g = 1; gems[g] != 0; g++) { char
temp[BUFLLEN]; strcpy(temp, name); char *token = strtok(temp, " "); while (token !=
NULL) { if (strcmp(token, gems[g]) == 0) { if (g > max_p) { max_p = g; } } token =
strtok(NULL, " "); } } return max_p; }

int strcasecmp_custom(const char *s1, const char *s2) { while (*s1 && *s2) { int diff =
tolower((unsigned char)*s1) - tolower((unsigned char)*s2); if (diff != 0) return diff; s1++;
s2++; } return tolower((unsigned char)*s1) - tolower((unsigned char)*s2); }

int main() { int n = 0; while (fgets(ponies[n], BUFLLEN, stdin))
{ ponies[n][strcspn(ponies[n], "\r\n")] = '\0'; if (strcmp(ponies[n], "END") == 0) break; if
(strlen(ponies[n]) > 0) { gem_rank[n] = get_priority(ponies[n]); n++; } }

for (int a = 0; a < n - 1; a++) {
    for (int b = a + 1; b < n; b++) {
        int swap = 0;
        if (gem_rank[a] != gem_rank[b]) {
            if (gem_rank[a] < gem_rank[b]) swap = 1;
        } else {
            if (strcmp(ponies[a], ponies[b]) > 0) {
                swap = 1;
            }
        }
        if (swap) {
            char temp[BUFLLEN];
            strcpy(temp, ponies[a]);
            strcpy(ponies[a], ponies[b]);
            strcpy(ponies[b], temp);

            int tr = gem_rank[a];
            gem_rank[a] = gem_rank[b];
            gem_rank[b] = tr;
        }
    }
}
```

```
    }  
  }  
}  
  
for (int i = 0; i < n; i++) {  
    printf("%s\n", ponies[i]);  
}  
  
return 0;  
  
}
```

Input:

```
Buttershy  
Orangejack  
Rubyanne  
Rainbow Dash  
Rarity  
Pinkie Pie  
Emerald Sunshine  
Spike the Dragon  
Misty Sapphire  
END
```

Output:

```
Misty Sapphire  
  
Emerald Sunshine  
  
Buttershy  
  
Orangejack  
  
Pinkie Pie
```

Rainbow Dash

Rarity

Rubyanne

Spike the Dragon